Advanced Power BI Setup Guide: From Blank File to Advanced Interactivity

This guide walks you through every step of the Apex Retail Group project in the correct sequence. It is not a substitute for the project brief. Use them together. This guide focuses on the mechanics of building features in Power BI Desktop. The brief tells you what to build and why.

Files needed before you start:

* apex_stores.csv, apex_categories.csv, apex_sales.csv, apex_targets.csv
* Power BI Desktop installed (free download from powerbi.microsoft.com)
* Save all four CSV files into one folder. Name it DACP_Advanced_PowerBI.
* Create a blank Word or Notepad file alongside it called Verification Notes. You will log your cross-checks here.

PART 1 * Load Data and Build the Model

Step 1: Load all four files into Power Query

Open Power BI Desktop. Click Home > Get Data > Text/CSV. Load apex_stores.csv first. When the preview appears, click Transform Data - never click Load directly. Repeat for each file. You should have four queries in the Queries panel before doing anything else.

Apply these type changes in Power Query for each table:

Table	Column	Change To
apex_stores (rename: Stores)	enterprise_zone	True/False logical type
apex_categories (rename: Categories)	gross_margin_pct	Decimal Number
apex_sales (rename: Sales)	sale_month	Date
apex_sales	year, month_num	Whole Number
apex_targets (rename: Targets)	target_month	Date
apex_targets	revenue_target_gbp	Decimal Number

After all changes are applied, click Close and Apply.

Step 2: Create the DateTable

Go to Home > New Table. Paste the DateTable DAX from Section B of the project brief. After it loads, right-click DateTable in the Fields pane and select Mark as date table. Choose the Date column. If you skip this step, all time intelligence measures will return incorrect results.

Step 3: Build the five relationships in Model view

Switch to Model view. Drag to create each relationship below. Double-click each relationship line after creating it to verify the settings.

From (many side)	To (one side)	Cardinality	Cross-Filter
Sales[sale_month]	DateTable[Date]	Many to One	Single
Targets[target_month]	DateTable[Date]	Many to One	Single
Sales[store_id]	Stores[store_id]	Many to One	Single
Targets[store_id]	Stores[store_id]	Many to One	Single
Sales[category_id]	Categories[category_id]	Many to One	Single

You now have two fact tables (Sales and Targets) both connecting to Stores. This is correct star schema design, not an error. The relationships are independent. When you write measures using Targets data in a Sales context, use USERELATIONSHIP to activate the correct path.

Step 4: Create the Measures table

Go to Home > Enter Data. Create a table with one column called Placeholder and one blank row. Name it Measures. Click Load. In Data view, delete the Placeholder column. All your measures go in this table - never mix measures with data columns.

Step 5: Write and verify the foundation measures

Open the Measures table in Data view. Go to Table tools > New Measure. Write each foundation measure from Section C of the brief. After each measure, verify it immediately using the method below.

Verification method for every measure:

* Create a Matrix visual on a blank canvas page
* Add Stores[region] to the Rows well
* Add your new measure to the Values well
* Open your apex_sales.csv in Excel. Run a SUMIF for South East stores. Compare to the matrix.
* If they match: write 'Revenue - South East verified' in your Verification Notes file.
* If they do not match: check your relationship in Model view before continuing.

PART 2 * Time Intelligence Measures

Step 6: Write and test Revenue SPLY

After writing Revenue SPLY from Section D, create a line chart to test it. Add DateTable[Month Year] to the X axis. Add both Total Revenue and Revenue SPLY to the Y axis. The Revenue SPLY line should be blank for all of 2023 and should show 2023 values shifted forward by exactly 12 months in 2024.

If Revenue SPLY shows the same values as Total Revenue for every month, your DateTable is not marked as a date table. Go to the Fields pane, right-click DateTable, and check the Mark as date table setting. The calendar icon should appear next to DateTable when it is correctly marked.

Step 7: Test the Rolling 3-Month Average

Add Revenue Rolling 3M to your test line chart. The rolling average line should be smoother than the monthly revenue line - it should not spike as sharply in November and December. If it spikes as much as the monthly line, your DATESINPERIOD formula is incorrect. Check that you are referencing DateTable[Date] and that the interval is MONTH not DAY.

Step 8: Verify Revenue Cumulative

Add Revenue Cumulative to the line chart. This line should always increase (or stay flat) from left to right. It should never decrease. If it decreases, you have a filter context issue. The ALL(DateTable) in the measure is supposed to remove the time filter and allow the LESS THAN OR EQUAL comparison to work. If ALL(DateTable) is missing from your formula, the measure will not accumulate correctly.

PART 3 * RANKX and Dynamic Features

Step 9: Test RANKX at company level

Write the Store Revenue Rank (Company) measure. Create a Table visual with Stores[store_name] and the rank measure. Sort the table by the rank column ascending. Confirm the top-ranked store has rank 1. Confirm the total row shows blank (RANKX should return blank at the total level). If the total row shows a number, your measure is missing the IF(ISBLANK(vRevenue), BLANK(), RANKX(...)) wrapper.

Step 10: Test RANKX at region level

Write Store Revenue Rank (Region). Add Stores[region] to your test table alongside store name and both rank measures. A store ranked 5th in the company might be ranked 2nd in its region if it is in a smaller region. The region ranks should reset for each region - you should see ranks 1, 2, 3 appearing multiple times in different regions. If region ranks are the same as company ranks, your ALLEXCEPT is not working correctly. Check you are using ALLEXCEPT(Stores, Stores[region]) and not ALL(Stores).

Step 11: Create the Field Parameter

Go to Modelling > New Parameter > Fields. Name it 'Performance Metric Parameter'. Add these measures to the parameter: Total Revenue, Gross Profit, Gross Margin %, Revenue MoM % Change. Power BI will automatically create a table and a slicer. Place the slicer on Page 2 of your report. Connect a bar chart to the parameter by dragging the parameter's value field to the Y axis - the chart will then change based on the slicer selection.

Step 12: Write the Dynamic Chart Title

After creating the field parameter, write the Dynamic Chart Title measure from Section E. Apply it to a chart by going to Format pane > Title > Text. Click the fx button (conditional formatting). In the dialogue, set Field value and select Dynamic Chart Title from your Measures table. The title will now update automatically when the slicer changes or a region filter is applied.

PART 4 * Advanced Interactivity

Step 13: Build the Drill-Through page

Add a new report page. Name it 'Store Detail (Drill-Through)'. In the Filters pane on the right, look for the Drill-Through section (it may be collapsed). Drag Stores[store_name] into the Drill-Through well.

Build the required visuals on this page. Add a Back button via Insert > Buttons > Back. To test: go to Page 2, right-click any store name in a visual, and select Drill Through > Store Detail.

> Common drill-through problem: the drill-through option does not appear in the right-click menu. This usually means the dimension field (store_name) is not in the visual you are right-clicking on. The field used for drill-through must be present in the visual, not just in the Drill-Through filter well.

Step 14: Set up Bookmarks for the Before/After view

This step prepares for Section H. Go to View > Bookmarks. The Bookmarks pane will open on the left. Click Add bookmark. Name it 'Standard View'. Now change your report to show the EZ-adjusted measures (once you have written them in Section H). Click Add bookmark again. Name it 'EZ Adjusted View'. Insert two buttons (Insert > Buttons > Blank). Select a button. In the Format pane, go to Action. Set Type = Bookmark. Select the appropriate bookmark for each button. Apply contrasting labels so users know which state they are in.

Step 15: Build the Report Page Tooltip

Add a new report page. Name it 'Store Tooltip'. In the Format pane for this page, find the Page information section and toggle 'Allow use as tooltip' to On. Change the canvas size to Tooltip. Build a line chart showing 12-month revenue trend on this page. Go back to your main store ranking visual on Page 2. In the Format pane for that visual, go to Tooltips. Change Type from Default to Report page. Select Store Tooltip from the dropdown. Hover over a data point on Page 2 to test.

Step 16: Set up the What-If Parameter

Go to Modelling > New Parameter > Numeric Range. Name it 'Margin Improvement %'. Set Minimum = 0, Maximum = 10, Increment = 0.5, Default = 0. Write the Adjusted Gross Margin % measure from Section F. Add the parameter slicer to Page 3. The slicer default value of 0 means no adjustment - moving the slider to 2.5 simulates a 2.5 percentage point margin improvement across all stores.

PART 5 * The Enterprise Zone Measures (Section H)

Step 17: Write and test EZ Adjusted Cost

Write the EZ Adjusted Cost measure using the SUMX pattern from Section H. To test it: create a table visual with Stores[enterprise_zone] on rows and two columns: Total Cost and EZ Adjusted Cost. For rows where enterprise_zone = FALSE, both measures should show the same value. For rows where enterprise_zone = TRUE, EZ Adjusted Cost should be lower by 14 to 22 percent depending on the store's ez_tax_relief_pct. If both values are identical for EZ stores, check your RELATED function is pulling from the correct column.

Step 18: Verify EZ adjusted margins change the right stores

Build a table with store_name, Gross Margin % (unadjusted), and EZ Adjusted Gross Margin %. Filter to Enterprise Zone stores only. All six EZ stores should show a higher margin in the adjusted column. Non-EZ stores should show identical values in both columns. Run this check before writing the EZ ranking measures - you cannot rank on a broken measure.

Step 19: Apply conditional formatting to the Before/After table

On the Enterprise Zone page, build the comparison table from Section H Step 3. To apply conditional formatting: select the rank change column in the visual. Go to the Format pane > Cell elements > Background color. Toggle it On. Click the fx button. Set Format style to Rules. Rule 1: If value is greater than 0, colour = green. Rule 2: If value is less than 0, colour = red. Rule 3: If value equals 0, colour = grey.

PART 6 * Power BI Copilot

Step 20: Access Copilot (or use the alternative)

Copilot requires a Microsoft Fabric or Power BI Premium per-user licence. If you have access, open the Copilot pane from the Home ribbon in Power BI Desktop or Service. If you do not have access, use Power BI Q and A for Tasks 1 and 2: click the Q and A visual in the Visualisations pane, or use the Q and A icon on a published report in Power BI Service. For Task 3 (measure explanation), use ChatGPT as an alternative.

Step 21: Evaluate every Copilot output critically

After receiving any Copilot-generated output, do not accept it without checking. Follow this sequence:

* Check the measure used: hover over each visual to see which measure is powering it. Is it one of your verified measures?
* Check the numbers: do they match your verified matrix visual? If Copilot generated a new calculation, verify it manually.
* Check the chart type: would you have made the same choice? Why or why not?
* Check for missing business context: does the output serve the Commercial Director's question, or is it generic?

PART 7 * RLS and Publishing

Step 22: Create the six RLS roles

Go to Modelling > Manage Roles > Create. Build five regional manager roles and one Commercial Director role. The Commercial Director role has no filter applied. For each regional role, apply [region] = "South East" (and the appropriate region name for each). Save after all six roles are created.

Step 23: Test RLS using View As

Go to Modelling > View As Roles. Test the South East Manager role first. Confirm that: the store count KPI shows only South East stores, the region bar chart shows only South East, all measures reflect South East data only. Switch to Commercial Director role and confirm all 60 stores are visible. Test at least two roles and screenshot both for your submission. Click Stop Viewing when done.

Step 24: Save and publish

Go to File > Save As. Name your file DACP_Advanced_PowerBI_Apex.pbix. Publish via Home > Publish. Select My workspace. Verify the report opens correctly in Power BI Service at app.powerbi.com. Assign RLS roles in Service: go to the dataset > More options > Security.

Final check before submitting:

* Open your .pbix file on a different machine or re-open it. Check that all measures still work and that the model relationships are intact. If a measure shows an error after reopening, the most common cause is a reference to a table or column that was renamed.